

Jin-Song Dong, FIEAus

(Australian citizen)

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Bio:

Jin-Song Dong is a full professor at the National University of Singapore. His research spans a range of fields, including formal methods with LLM agents, safety and security systems, trusted AI, probabilistic reasoning, sports analytics, and verified LLM code synthesis. He co-founded the commercialized PAT verification system, which has garnered thousands of registered users from over 150 countries. Jin Song also co-founded the commercialized trusted machine learning system Silas (www.depintel.com), with 50K+ downloads. He has published 200+ papers in top journals/conferences (i.e., ICML, NeurIPS, ICLR, CVPR, IJCAI, AAAI, POPL, ICSE, FM, CAV, etc), and received numerous best paper awards. He served on the editorial board of ACM Transactions on Software Engineering and Methodology, Formal Aspects of Computing, and Innovations in Systems and Software Engineering, A NASA Journal. Jin Song has been General/Program Chairs for many international conferences, including FM 2014 when FM went to Asia for the first time. He has successfully supervised 34 PhD students, many of whom have become tenured faculty members at leading universities worldwide. He is a Fellow of the Institute of Engineers Australia. Jin Song developed Markov Decision Process (MDP) models for tennis strategy analysis using PAT, assisting professional players with pre-match analysis (beating the world's best). He founded a new international conference series on sports analytics (ISACE). In his spare time, he is a tennis coach and takes pleasure in coaching tennis to his students and his three children, all of whom have reached the #1 national junior ranking in Singapore/Australia. Two of his children have earned US NCAA Division-1 full scholarships.

Education

- PhD, [The University of Queensland, Australia](#) (1993 – 1995)
- BInfTech with 1st Class Honours (in Software Engineering), The University of Queensland, Australia (1989 - 1992)

Professional Work Experiences

- Full Professor (2016 -), Associate Professor (tenured) (2005 - 2016), Assistant Professor (1998-2004)
Computer Science Department (CS), School of Computing (SoC), National University of Singapore (NUS)
 - Graduate Committee (SoC, 2001-2014, 2022-current).
 - NUS Senate Member (2020-current)
 - Affiliated Professor and Master Project supervisor at NUS Business Analytic Center (2014 – current)
 - Member of PhD supervisors at [NUS Graduate School for Integrative Sciences and Engineering \(NGS\)](#)
 - CS Exco Committee Member (CS Department, 2008-2011, 2020-2024).
 - Faculty Promotion Committee (SoC, 2021-2024).
 - Deputy Head (CS Department, 2023-2024).
 - Search Committee (CS Department, 2005-2008, 2019-2023).
 - Co-Director of the French-Singapore joint [IPAL Lab](#) (2011 – 2019)
 - PhD Qualify Exam Coordinator (CS Department, 2008-2014).
 - Assistant Dean (Graduate Office, SoC, 2005-2008, 2022).
 - *Taught various undergraduate and postgraduate courses at SoC covering topics, i.e., Software Engineering, UML, Formal Methods, Concurrent Systems, Model Checking, Probabilistic Reasoning, Web Ontology, Discrete Maths etc*
 - *Taught various short courses on verification technology for software safety and security for government agency and engineering companies, i.e., DSO, DSTA, ST Engineering, Denso, Mitsubishi, NTT etc*
- Visiting Professor, National Institute of Informatics of Japan (2008-2017)
- Visiting Fellow at Kellogg College, Software Engineering Center, Oxford University UK (elected in 2006)
- Senior Research Engineer (1998), Research Engineering (1995-98), Software Engineering Group, [Commonwealth Scientific and Industrial Research Organisation \(CSIRO\)](#), Australia

Commercial Company and Systems:

- Co-founded (with two former Phd Students Jun/Yang) Semantic Engineering PTE LTD and the advanced PAT model checking system (~1 million lines source code) which has **4000+** registered users from **1000+** organizations in **150 countries**, including Toyota, NTT, Mitsubishi, JAXA, etc. PAT won “20 Year ICFEM Most Influential System Award” in 2018.
- Co-founded (with two former postdocs, Zhe/Hadrien) Dependable Intelligence PTE LTD and Silas Machine Learning system (with **50K+ download**) that has explainable and verifiable capabilities. It has commercial applications in Health, Environment and Sports Analytics. www.depintel.com

Awards (Selected):

- First Prize at Research Prototype System Competition, CCF ChinaSoft 2025 (top 0.1%)
- Singapore Cyber Grand Challenge Award 2024
- ACM SIGSOFT Distinguished Paper Award for ICSE 2020
- NUS Research Recognition Award (2020, 1-3 selected from university wide all senior professors)
- Fellow of Institute of Engineers Australia (2018)
- 20 Year ICFEM Most Influential System Award (2018)
- Best Paper Award at Software and Systems Modeling (SSM) journal (2015)
- Visiting Fellow at Kellogg College, Oxford University (elected during 2006 sabbatical)
- NUS Young Researcher Award (2004, 1-4 selected from university wide all junior professors)
- Richard Jago Memorial Prize (1995, best computing PhD student) and Australian Postgraduate Award (93-95)

Professional and Research Activities (selected)

- Member of Singapore Government Coordinating Committee for Artificial Intelligence (2024-2026) (one from NUS)
- Editorial Board Member of ACM Transaction on Software Engineering & Methodology (TOSEM) (2012-2020)
- Editorial Board Member of [Formal Aspects of Computing Journal](#) (FAC)
- Editorial Board Member of [Innovations in Systems and Software Engineering, A NASA Journal](#) (ISSE)
- Advisory Board Member of [Formal Methods Europe \(FME\)](#)
- Steering Committee Member of IEEE Pacific Rim International Symposium on Dependable Computing
- Steering Committee Member of International Conference on Engineering of Complex Computer Systems
- Steering Committee Chair of [International Conference on Formal Engineering Methods](#) (ICFEM)
- Steering Committee Member of Asia-Pacific Software Engineering Conference (APSEC)
- General Chair of FM 2014 Singapore (first time FM goes to Asia) and General/PC Chairs of many other conferences
- Founder and Conference Series Chair of International Sports Analytics Conference and Exhibition (ISACE)
- PC members of international conferences, i.e., ICSE, ISSTA, FM, Nasa-FM, ASE, etc
- External Funding Evaluator
 - Australian Research Council (2008, 2014, 2020-2025)
 - ARC ERA REC Panel Member (2018)
 - Swiss National Science Foundation (2017-2018)
 - South Africa's National Research Foundation (2015)
 - Expert panel for MND-ECAC joint research fund, Building and Construction Authority, Singapore (2015)
 - Natural Sciences and Engineering Research Council of Canada (2014)
 - Romania Executive Agency for Higher Education, Research, Development and Innovation (2012, 2013)

Selected Research projects (Principal Investigator over 25 million USD individual projects), converted to USD:

- Secure, Private, and Verified Data Sharing for Large Model Training and Deployment, Lead PI, \$5M, 2024-2029, NRF Cyber and AI Singapore
- Towards Building Unified Autonomous Vehicle Scene Representation for Physical AV Adversarial Attacks and Visual Robustness Enhancement, Co-PI, \$700K, AI Singapore, 2025-2026
- Testing and Formal Analysis of the Robustness of Perception Systems for Autonomous Driving, NUS-RI Chongqing, supported by ChangAn company, \$200K, 2024-2026.
- Visualizing and Debugging Training Dynamics of Deep Learning Models via Time-Travelling Visualization, Lead PI, 2025-2028 \$470K, MOE Tier2
- Enhancing Privacy Regulation Compliance among Web-based Applications, Lead PI, 2025-2028, \$508K, MOE Tier2
- Enhancing Large Language Models with Rigorous Reasoning, Lead PI, 2024-2026 \$200K, NUS AI Institute
- A Pre-trained Multimodal Commonsense Transformer for Human Omnitraining, Co-PI, \$4M, 2024-29, AI Singapore
- An AI-based Anti-Phishing Platform for Monitoring and Analysing Runtime Phishing Campaigns (\$400K, 2022-2024) NRF Cyber Security Grant

- Trustworthy AI Centre NTU (individual: \$342K, 2023-2026)
- CNRS-IMT French Funding: Formal Methods for AAL (\$550K, 2015-2024)
- MOE T2 An Execution Trace Alignment Based Approach to Locating Regression Bug with Explanation (\$448K, 2021-2024)
- MOE T3 The Science of Certified AI Systems (total \$6M, individual: \$800K, 2021-2026)
- Chief Investigator, ARC Industry Transformation Hub on AI for Farming (\$7M, 2018-2023)
- Lead Chief Investigator, Nested Graph transformations and Trusted Autonomy, DST grants (\$350K, 2019-2021)
- Chief Investigator, Australian Aged Care Technologies Collaborative (AACTC) and Event Analytics (\$200K, 2019-2022).
- Chief Investigator, Towards developing Indo-Australian partnership for digital transformation through blockchain (\$100K, 2018-2019)
- Principal Investigator of “Predicative Security Analytics” in \$38M SingTel-NUS Joint Cyber Security Lab (individual component \$3M, 2016-2025)
- Partner Investigator of European Commission Horizon 2020 programme on “Elderly-friendly City services for active and healthy ageing” (\$5M, IPAL(CNRS) partner component is \$600K with Mounir Mokhtari, 2015-2018)
- NRF-Cyber Security Program on " *Trustworthy systems from untrusted Components* " (individual component \$600K, 2014-2019)
- Singapore Principal Investigator of Singapore-UK Cyber Security Project on “*Security and Privacy in Smart Grid Systems: Countermeasure and Formal Verification*” (with Oxford University), (\$600K, 2016 – 2018)
- Principal Investigator of French-FRC joint research project “*Reliable Prototyping Framework for Daily Living Assistance of Frail Ageing People (RELIANCE)*” (\$200K, 2014 - 2017)
- Principal Investigator of A*STAR PSF “*Formal Verification Systems for Simulink/Stateflow*” (\$400K, 2012 - 2015)
- Co-Principal Investigator of Tlab “*Formal Verification for Cyber Security*” (\$720K, 2012 - 2014)
- Principal Investigator of MOE (Tier 2) project “*Advanced Model Checking Systems*” (\$718K, 2009 - 2012)
- Principal Investigator of NRF MDA project “*Systematic Design Methods and Tools for Developing Location Aware, Mobile and Pervasive Computing Systems*” (\$311K, 2007-2010)
- Principal Investigator of MOE (Tier 2) project “*Rigorous Design Methods and Tools for Intelligent Autonomous Multi-Agent Systems*” (\$500K, 2004-2008)
- Principal Investigator of DIRP (DSTA) project “*Formal Design Methods and DAML*” (\$314K, 2002 – 2004)
- Principal Investigator of the US Airforce Research Lab fund “*Timed Formalisms for Plan Ontology*” (\$62K 2006-08)

Current PhD students (8):

Rajdeep Hundal, Sun Changsheng, Wang Hongshu, Zuo Xinyue, Ma Murong, Gao Tianyu, Liu Chenyan, Teoh Xiwen

Supervised PhD students (34) at NUS

1. Liu Yang PhD, -> Full Chair Professor at Nanyang Technological University
2. Sun Jun PhD, -> Full Professor at Singapore Management University
3. Yuan Ling PhD, -> Full Professor at Huazhong University of Science and Technology
4. Li Yuanfang PhD, -> Associate Professor at Monash University, Australia, -> Chief AI Scientist, Oracle.
5. Bai Guangdong PhD, -> Associate Professor, University of Queensland
6. Liu Shuang PhD, -> Associate Professor, Renmin University
7. Sun Jing PhD, -> Associate Professor, The University of Auckland, NZ
8. Wong Kailong PhD, -> Associate Professor at Huazhong University of Science and Technology
9. Wang Hai PhD, -> Senior Lecturer at Aston University, UK
10. Kulani Tharaka Mahadewa, PhD -> Senior Lecturer, University of Moratuwa
11. Mark Huasong Meng, PhD -> Assistant Professor at University College Dublin, Ireland
12. Ye Quanqi PhD, -> Lecturer at Waikato University
13. Jiang Kan PhD, -> Lecturer, National University of Singapore
14. Dileepa PhD, -> Lecturer, Singapore University of Technology and Design
15. Liang Hui PhD, -> Lead Data Scientist, Innovation Center, HP Enterprise
16. Feng Yuzhang PhD, -> First Vice President (Analytics Strategy) at United Overseas Bank Limited (UOB)
17. Zhang Shaojie PhD, -> Software Engineer, Google
18. Zheng Manchun PhD, -> Member of Technical Staff at Pure Storage, USA
19. Tan Tian Huat PhD, -> Data Scientist at IBM
20. Gui Lin PhD, -> Vice President of Data Science at Lazada Group
21. Li Li PhD, -> Engineering Manager at Apple
22. HAO Ping, PhD, -> Tech Lead, Prudential Service SGP

23. CHEN Chunqing, PhD -> Manager of Smart City Program, Infocomm Development Authority of Singapore
24. ZHANG Xian, PhD -> Postdoc at I2R-NUS Research Project
25. SONG SongZheng, PhD -> Research Fellow at NTU
26. ZHU Huiquan, PhD -> Research Fellow at NUS
27. LIU Yan, PhD -> Postdoc at Advanced Digital Sciences Center
28. SHI Ling, PhD -> Research Manager at NTU
29. NGUYEN Truong Khanh, PhD -> Principal Engineer, Atlassian
30. Chen Manman, PhD -> Director of Machine Learning and GenAI, American Express
31. Cai Yunfan, PhD -> RF, NUS
32. Yang Xianglin, PhD -> RF, NUS
33. Liu Ruofan, PhD -> RF, NUS
34. Liu Zhaoyu, PhD -> RF, NUS

Completed research MSc students (10) and course MComp project students (50+)

JI Kun, ZHANG Jiexin, LUU Anh Tuan (-> Assistant Professor, NTU), WANG Xiaohang (-> Google NYC Lab), Huang Xiaoning, TANG Yue Linda, WANG Yan, LIU Jing, ONG Wai Chun West Jean, TEH Hsin Yee

Supervised postdocs and research fellows (27)

Xu Ming, Qi Binhang, Liu Zhaoyu, Yang Xianlin, Cai Yufan, Zhang Hangsheng, Zhang Yeding, Jiang Kan, Lin Yun, Naipend Dong, Etienne Andrew, LIN Shangwei, SHI Jiaqi, SHAO Rongxin, CHEN Tieming, ZHANG Xian, CHEN Chunqing, SUN Jun, WANG Hui, LI Yuanfang, TUN Nwe Ni, YUAN Ling, LIANG Hui, SUN Jing, YANG Shaofa, Bai Guangdong, Shi Ling, Cyrille Jegourel, HOU Zhe, Hadrien BRIDE.

Selected publications in recent years (total 300+ publications with 13000+ citations: [google scholar](#))

2026

1. Y Liao, Y Cao, Y Zhang, W He, Y Xiao, X Du, Z Huang, J Dong, Towards Stealthy and Effective Backdoor Attacks on Lane Detection: A Naturalistic Data Poisoning Approach. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026
2. M Ma, R Liu, Y Lin, Z Huang, J Dong. TrainRef: Curating Data with Label Distribution and Minimal Reference for Accurate Prediction and Reliable Confidence. ICLR 2026
3. Y Cao, W Song, D Wang, J Xue, J Dong. Failures to Surface Harmful Contents in Video Large Language Models, AAAI 2026.
4. Z Liu, K Jiang, M Ma, Z Hou, Y Lin, J Dong. Few-Shot Precise Event Spotting via Unified Multi-Entity Graph and Distillation, AAAI 2026.
5. H Wang, M Xu, Y Guo, W Han, H Lim, J Dong. RulePilot: An LLM-Powered Agent for Security Rule Generation, ICSE 2026.
6. Z Zhao, J Sun, Z Hou, Z Wei, C Cai, M Qiao, J Dong. MaCTG: Multi-Agent Collaborative Thought Graph for Automatic Programming, ICSE 2026.
7. W Zhang, Y Lin, K Fung, X Teoh, X Xie, F Liauw, H Zhang, J Dong. MINES: Explainable Anomaly Detection through Web API Invariant Inference. ICSE 2026
8. J Hu, J Sun, C Wang, Y Huang, J Feng, Y Xu, Y Li, K Wang, W Miao, J Dong, G Pu. ADARULE: LLM-Driven Natural Language to LTL Conversion via Pattern-Adaptive Rule Induction, ICSE 2026.
9. H Wang, X Zuo, Y Sun, Q Li, Y Ameur, J Dong. Towards LLM Agent for Formal Model Synthesis and Fixing, FSE, 2026
10. X Teoh, Y Lin, D Nguyen, R Ren, W Zhang, J Dong. WebTestPilot: Agentic End-to-End Web Testing against Natural Language Specification by Inferring Oracles with Symbolized GUI Elements. (FSE 2026).
11. B Qi, Y Lin, X Weng, C Liu, H Sun, G Fraser, J Dong. Generalizing Test Cases for Comprehensive Test Scenario Coverage. FSE 2026.
12. X Yang, B Hooi, G Deng, T Zhang, J Dong. Turning Bias into Bugs: Bandit-Guided Style Manipulation Attacks on LLM Judges. ICML 2026.
13. X Lu, X Yang, H Liu, J Liu, K Cai, Y Xiao, J Dong. Inverting the Shield: Systematically Generating Safety Tests from Policy Specifications. ACL 2026.
14. X Ma, Y Zhou, W Sun, Z Liu, J Leng, Y Lin, S Sun, M Guo, J Dong. CuBridge: An LLM-Based Framework for Understanding and Reconstructing High-Performance Attention Kernels. ACL 2026.
15. Y Xiao; Y Chen; J Wang; W Song; J Sun; S Ma; Y Mu; J Zhai; C Fang; J Dong; Z Chen. Train in Vain: Functionality-Preserving Poisoning to Prevent Unauthorized Use of Code Datasets. ACL Findings 2026
16. M Xu, H Wang, Y Guo, Z Yu, W Han, H Lim, J Dong, J Zhang, ARuleCon: Agentic Security Rule Conversion, WWW 2026
17. X Chen, P Zhang, J Sun, J Dong, W Wang, J Wang, LLMQuA: Practical Backdoor Injection on Large Language Model Quantization. WWW 2026.
18. Z Xu, K Sima, J Deng, Z Zhuang, C Wang, C Hao, J Dong. APREBot: Active Perception System for Reflexive Evasion Robot. IEEE International Conference on Robotics and Automation (ICRA). 2026

19. Z. Xu, C Hao, C Wang, K. Sima, F Shi, J Dong. REBot: Reflexive Evasion Robot for Instantaneous Dynamic Obstacle Avoidance. IEEE International Conference on Robotics and Automation (ICRA). 2026
20. Y Dong; X Chen; X Song; Y Yang; W Lu; T Zhang; J Zhou; J Dong: Alkaid: Accelerating Three-Party Boolean Circuits by Mixing Correlations and Redundancy. IEEE Transactions on Information Forensics and Security. 2026
21. Y Cao; K Ye; D Wang; M Xue; H Ge; C Qian; J Dong: Bones of Contention: Exploring Query-Efficient Attacks Against Skeleton Recognition Systems. IEEE Transactions on Information Forensics and Security. 2026
22. Z Xu; C Hao; C Wang; K Sima; F Shi; J Dong: REBot: Reflexive Evasion Robot for Instantaneous Dynamic Obstacle Avoidance. IEEE Robotics and Automation Letters. 2026.
23. T Mead, Z Wang, E Foo, J Dong, N Dong, R Ko, A Koay, K Nguyen, Multi-agent reinforcement curriculum learning for real unmanned ground vehicles, Engineering Applications of Artificial Intelligence 167, 113780, 2026
24. H Meng, C Yan, Q Zhang, Z Wang, K Wang, S Teo, G Bai, J Dong, Assessing Privacy Disclosure Compliance of Android Third-Party SDK, IEEE Transactions on Software Engineering, 2026

2025

25. Y Zhang, Y Sun, A Lee, J Dong, RvLLM: LLM Runtime Verification with Domain Knowledge, NeurIPS 2025
26. W Jin, Y Cao, J Su, J Xue, J Hao, K Xu, J Dong, D Wang, ALMGuard: Safety Shortcuts and Where to Find Them as Guardrails for Audio-Language Models, NeurIPS 2025
27. W Dai, B Li, N Dong, G Bai, J Dong, FracFace: Breaking the Visual Clues—Fractal-Based Privacy-Preserving Face Recognition, NeurIPS 2025
28. Y Zhang, Y Cai, X Zuo, X Luan, K Wang, M Sun, Z Hou, J Sun, J Dong, Position: Trustworthy AI Agents Require the Integration of LLM with Formal Methods, ICML 2025
29. X Weng, J An, X Ma, B Qi, J Luo, X Yang, J Dong, L Huang, Clustering Properties of Self-Supervised Learning, ICML 2025
30. Q Zhou, T Li, Q Guo, D Wang, Y Lin, Y Liu, J Dong, Defending LVLMs Against Vision Attacks through Partial-Perception Supervision, ICML 2025
31. Y Cai, Z Hou, D Sanan, X Luan, Y Lin, J Sun, J Dong. Automated Program Refinement: Guide and Verify Code Large Language Model with Refinement Calculus. POPL 2025.
32. Z Liu, J Kun, Z Hou, J Dong, F3Set Towards Analyzing Fast, Frequent, and Fine-grained Events from Videos, ICLR, 2025
33. X Zuo, Y Zhang, H Wang, Y Cai, Z Hou, J Sun, J Dong. PAT-Agent: Autoformalization for Model Checking. IEEE/ACM International Conference on Automated Software Engineering (ASE) 2025.
34. C Liu, Y Lin, Y Huang, J Chang, B Qi, B Jiang, Z Huang, J Dong. Learning Project-wise Subsequent Code Edits via Interleaving Neural-based Induction and Tool-based Deduction. (ASE 2025).
35. W Jin, Y Cao, J Su, D Wang, Y Zhang, M Xue, J Hao, J Dong, Y Yang, Whispering Under the Eaves: Protecting User Privacy Against Commercial and LLM-powered Automatic Speech Recognition Systems, USENIX Security 2025.
36. X Teoh, Y Lin, S Li, R Liu, A Sollomoni, Y Harel, J Dong. Are CAPTCHAs Still Bot-hard? Generalized Visual CAPTCHA Solving with Agentic Vision Language Model. USENIX Security 2025.
37. R Hundal, Y Xiao, X Cao, J Dong, M Rigger. On the Mistaken Assumption of Interchangeable Deep Reinforcement Learning Implementations. ICSE 2025.
38. Y Zhang, L Huang, P Gao, F Song, J Sun, J Dong, Verification of Bit-Flip Attacks against Quantized Neural Networks, OOPSLA 2025
39. J Dong, J Sun, W Zhang, JS Dong, D Hao: Consistency-Aided Tested Code Generation with LLM, ISSTA 2025.
40. R Liu, X Teoh, Y Lin, D Poshyvanyk, J Dong: GUIPilot: A Consistency-based Mobile GUI Testing Approach for Detecting Application-specific Bugs, ISSTA 2025.
41. Y Zhou, H Zhu, Q Qiu, W Cui, Z Liu, P Chen, M Wahib, C Guo, S Feng, J Meng, H Lan, J Leng, Y Lin, J Dong, W Zhu, M Deng. A Sample-Free Compilation Framework for Efficient Dynamic Tensor Computation Proceedings of the International Conference for High Performance Computing, SC 2025
42. X Zuo, Y Xiao, X Cao, W Wang, J Dong. DT4LM: Differential Testing for Reliable Language Model Updates in Classification Tasks. IEEE Transactions on Software Engineering (TSE), 2025
43. H Zhang, D Han, S Zhuang, Z Wang, J Sun, Y Liu, J Liu, J Dong. Explainable and transferable adversarial attack for ml-based network intrusion detectors. IEEE Transactions on Dependable and Secure Computing, 2025.
44. G Bai, K Wang, Y Ling, Y Zhang, Z Yu, H Wang, B Ooi, J Dong: Characterizing Cryptocurrency-Themed Malicious Browser Extensions. Commun. ACM 68(11): 93-101 (2025)
45. Y Huang, C Liu, Y Lin, Y Cai, B Jiang, P Yang, Z Huang, J Dong: CoEdPilot: Interactively Recommending Project-Wise Code Edits. J. Comput. Sci. Technol. 40(4): 980-992 (2025)
46. Y Xiao, X Zuo, X Lu, J Dong, X Cao, I Beschastnikh: Promises and perils of using Transformer-based models for SE research. Neural Networks 184: 107067 (2025)

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47. Y Zhang, G Chen, F Song, J Sun, J Dong, Certified Quantization Strategy Synthesis for Neural Networks, International Symposium on Formal Methods (FM), 2024.
48. R Liu, Y Lin, X Teoh, Z Huang, J Dong. Less Defined Knowledge and More True Alarms: Reference-based Phishing Detection without a Pre-defined Reference List (USENIX Security 2024).
49. X Teoh, Y Lin, R Liu, Z Huang, J Dong. PhishDecloaker: Detecting CAPTCHA-cloaked Phishing Websites via Hybrid Vision-based Interactive Models (USENIX Security 2024).

50. C Liu, Y Cai, Y Lin, Y Huang, Y Pei, J Dong, H Mei. CoEdPilot: Recommending Code Edits with Learned Prior Edit Relevance, Project-wise Awareness, and Interactive Nature (ISSTA 2024).
51. Y Pan, J Liu, T Zhao, L Zhang, Y Lin, J Dong. A Symbolic Rule Integration Framework with Logic Transformer for Inductive Relation Prediction (WWW 2024).
52. J Liu, X Liao, J Dong, A Quaternion-Valued Neural Network Approach to Nonsmooth Nonconvex Constrained Optimization in Quaternion Domain, *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2023
53. J Liu, X Liao, J Dong, A Mansoori, A neurodynamic approach for nonsmooth optimal power consumption of intelligent and connected vehicles, *Neural Networks* 161, 693-707, 2023
54. J Liu, X Liao, J Dong, A Mansoori, A subgradient-based neurodynamic algorithm to constrained nonsmooth nonconvex interval-valued optimization. *Neural Networks* 160, 259-273, 2023
55. J Liu, X Liao, JS Dong, A Recurrent Neural Network Approach for Constrained Distributed Fuzzy Convex Optimization. *IEEE Transactions on Neural Networks and Learning Systems*. 2023
56. Y Cai, Y Lin, C Liu, J Wu, Y Liu, Y Gong, J Dong. On-the-Fly Adapting Code Summarization on Trainable Cost-Effective Language Models (NeurIPS 2023).
57. Z Liu, K Jiang, Z Hou, Y Lin, J Dong. Insight Analysis for Tennis Strategy and Tactics (ICDM 2023).
58. X Yang, Y Lin, Y Zhang, L Huang, J Dong, H Mei. An Interactive Time-Travelling Debugging Approach for Deep Classifiers (FSE 2023).
59. G Zhang, Z Hóu, Y Huang, J Shi, H Bride, J Dong, Y Gao: Extracting optimal explanations for ensemble trees via automated reasoning. *Appl. Intell.* 53(11): 14371-14382 (2023)
60. X Ren, Y Lin, Y Xue, R Liu, J Sun, Z Feng, J Dong: DeepArc: Modularizing Neural Networks for the Model Maintenance. *ICSE 2023*: 1008-1019
61. M Liu, Y Lin, J Liu, B Liu, Q Zheng, J Dong: B2-Sampling: Fusing Balanced and Biased Sampling for Graph Contrastive Learning. *KDD 2023*: 1489-1500
62. J. Wang, M. Luo, J. Li, Y. Lin, Y. Dong, J. S. Dong, Q. Zheng: Empower Post-hoc Graph Explanations with Information Bottleneck: A Pre-training and Fine-tuning Perspective. *KDD 2023*: 2349-2360
63. K. Wang, Y. Ling, Y. Zhang, Z. Yu, H. Wang, G. Bai, B. C. Ooi, J. S. Dong: Characterizing Cryptocurrency-themed Malicious Browser Extensions. *SIGMETRICS 2023*: 91-92
64. R. Liu, Y. Lin, Y. Zhang, P. Lee, J. S. Dong: Knowledge Expansion and Counterfactual Interaction for Reference-Based Phishing Detection. *USENIX Security Symposium 2023*: 4139-4156
65. H Meng, Q Zhang, G Xia, Y Zheng, Y Zhang, G Bai, Z Liu, S Teo, J Dong Post-GDPR Threat Hunting on Android Phones: Dissecting OS-level Safeguards of User-unresettable Identifiers *The Network and Distributed System Security Symposium (NDSS) 2023*.
66. R. Liu, Y. Lin, X. Yang, J. S. Dong. Debugging and Explaining Metric Learning Approaches: An Influence Function Based Perspective (oral, NeurIPS 2022).
67. K. Wang, Y. Zheng, Q. Zhang, G. Bai, M. Qin, D. Zhang, J. S. Dong Assessing Certificate Validation User Interfaces of WPA Supplicants *The 28th Annual International Conference On Mobile Computing And Networking (MobiCom)*, 2022.
68. Y. Ling, K. Wang, G. Bai, H. Wang, J. S. Dong Are They Toeing the Line? Diagnosing Privacy Compliance Violations among Browser Extensions *The 37th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2022.
69. Y Xiao, Y Lin, I Beschastnikh, C Sun, D Rosenblum, J Dong. Repairing Failure-inducing Inputs with Input Reflection (ASE 2022).
70. S. Li, X. Xie, Y. Lin, Y. Li, R. Feng, X. Li, W. Ge, J. Dong. Deep Learning for Coverage-Guided Fuzzing: How Far Are We? (*IEEE Trans. Dependable and Secure Computing, TDSC* 2022).
71. Y. Xiao, I. Beschastnikh, Y. Lin, R. S. Hundal, X. Xie, D. Rosenblum, J. S. Dong. Self-Checking Deep Neural Networks for Anomalies and Adversaries in Deployment (*TDSC* 2022).
72. H. Meng, G. Bai, S. G. Teo, Z. Hou, Y. Xiao, Y. Lin, J. S. Dong. Adversarial Robustness of Deep Neural Networks: A Survey from a Formal Verification Perspective (*TDSC* 2022).
73. X. Song, Y. Lin, S. H. Ng, Y. Wu, X. Peng, J. S. Dong, H. Mei. RegMiner: Towards Constructing a Large Regression Dataset from Code Evolution History (*ISSTA* 2022).
74. X. Yang, Y. Lin, R. Liu, J. S. Dong. Temporality Spatialization: A Scalable and Faithful Time-Travelling Visualization for Deep Classifier Training (*IJCAI* 2022).
75. R. Liu, Y. Lin, X. Yang, S. H. Ng, D. M. Divakaran, and J. S. Dong. Inferring Phishing Intention via Webpage Appearance and Dynamics: A Deep Vision Based Approach (*USENIX Security* 2022).
76. X. Yang, Y. Lin, R. Liu, Z. He, C. Wang, J. S. Dong, H. Mei: DeepVisualInsight: Time-Travelling Visualization for Spatio-Temporal Causality of Deep Classification Training. *AAAI* 2022: 5359-5366
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Keynote/Invited-talk/Tutorials:

1. J. S. Dong, **Distinguished Lecture: Reasoning Beyond LLM: Code, Design, Strategy, and Life after Superintelligence: The Player Era**, HK University (Shanghai Site), Feb 2026.
2. J. S. Dong, **Invited Talk: Trusted AI and Reasoning Beyond LLM**, AAI Workshop on Post-AI Formal Methods, Singapore, Jan 2026. <https://www.youtube.com/watch?v=AEV6u4fy4n4>
3. J. S. Dong, **Invited Talk: Neurosymbolic Agents and Reasoning Beyond LLM**, LLM Training Workshop, Guangzhou, Dec 2025
4. J. S. Dong, **Invited Talk: Model Checking Agents and Reasoning Beyond LLM**, 20th Huawei Testing Conference, Nanjing, Nov 2025

5. J. S. Dong, **Keynote: Trusted AI and Reasoning Beyond LLM**, The Second Eurasian Forum on AI & Data Science Baku, Azerbaijan, Oct 2025
6. J. S. Dong, **Keynote: Trusted AI and Reasoning Beyond LLM**, 21th International Conference on Intelligent Computing (ICIC 2025), July 2025, Ningbo.
7. J. S. Dong, **Invited Talk: Model Checking Agents and Reasoning Beyond LLM**, Huawei Strategy and Technology Workshop & Global Software Technology Summit July 2025. Hong Kong.
8. J. S. Dong, **Invited Lecture: Trusted AI and Reasoning Beyond LLM**, 21st International Summer School on Trustworthy Software, Shanghai, July 2025
9. J. S. Dong, **Keynote: Dependable Intelligence and Reasoning Beyond ChatGPT+ with an Application to Sports Analytics**, The 6th IEEE World Symposium on Software Engineering, Kyoto, Sep 2024
10. J. S. Dong, **Keynote: Dependable Intelligence and Reasoning Beyond LLM**, CCF Conference on Programing and AI, Dali, Dec 2024.
11. J. S. Dong, **Distinguished Talk: Trusted AI for Sports Analytics**. Systems Engineering Workshop for Complex Intelligent Systems, May, 2024, Hong Kong Science Park. Chaired by M. Xie and J. Sifakis (2007 Turing Award)
12. J. S. Dong, **Keynote: Trusted AI and Reasoning beyond ChatGPT**, International Multi-Conference on "Advances in Information Technology, Communication and Energy Systems" (AITCES), Bengaluru, India, January 2024.
13. J. S. Dong, **Keynote: Probabilistic Reasoning for Sports Analytics**, 27th ICECCS, Toulouse, France, June 2023
14. J. S. Dong, **Keynote: Probabilistic Model Checking for Sports Strategy Analytics**, SETTA. Nanjing, 2023
15. J. S. Dong, **Keynote: Trusted Decision Making and Dependable Intelligence**, 19th International Conference On Smart Living and Public Health (ICOST), Paris, France, June 2022
16. J. S. Dong, **Keynote: Trusted Decision Making**, ICNSC 2020 : IEEE International Conference on Networking, Sensing and Control, 2020.
17. J. S. Dong, **Keynote: Trusted Decision Making and Dependable Intelligence**, AI-Expo, Suzhou, China, May, 2019
18. J. S. Dong, **Keynote: Decision Making via Model Checking**, Smart Digital Futures SDF-18, June, 2018
19. J. S. Dong, **Invited Talk: Event-Strategy Analytics**, Mathworks Summit, Tokyo, Oct, 2017
20. J. S. Dong, **Keynote: Event Analytics**, Australian System Safety Conference, Sydney, May 2017
21. J. S. Dong, **Invited Annual Lecture: Model Checking and Event Analytics**, The Engineering Australia Annual ITEE/IET Informatics Lecture, Brisbane, March 2017.
22. J. S. Dong, **Keynote: Event-Strategy Analytics**, 11th The 9th International Symposium on Theoretical Aspects of Software Engineering (TASE'15), September 12-14, 2015, Nanjing, China
23. J. S. Dong, **Invited Tutorial: Pervasive Model Checking**, 11th The 9th International Symposium on Theoretical Aspects of Software Engineering (TASE'15), September 12-14, 2015, Nanjing, China
24. J. S. Dong, **Keynote: Event Analytics**, 11th International Colloquium on Theoretical Aspects of Computing (ICTAC'14), 17-20 Sep 2014, Bucharest, Romania.
25. J. S. Dong, **Tutorial: Model Checking: PAT Approach**, 11th International Colloquium on Theoretical Aspects of Computing, 17 Sep 2014, Bucharest, Romania.
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ard in 2018 (with Jun and Yang). He also co-founded "Silas: Trusted Machine Learning" and the Dependable I